

Translating clinical narratives into structured evidence

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Translating clinical narratives into structured evidence with neuro-symbolic systems

Abstract—Clinical information extraction (IE) converts clinical narratives into structured facts. Large language models can do this with significantly less preprocessing than rules-based methods, but they introduce new classes of problems: hallucination, schema drift and overconfidence. We propose a hybrid system where the model extracts entities, interprets relations and provides evidence, and the rules normalise entities, verify evidence and apply policies to resolve conflicts. This modular system preserves the rich uncertainty captured in clinical narratives while producing auditable structured outputs that confirm to explicit rules. We use 2 public datasets of epilepsy letters to evaluate this system on two distinct tasks: (1) selecting the most important clinical fact among many and (2) building a broad inventory of clinical facts. We show that the combined system improved F1 by +0.05 vs. rules-only or LLM-only systems, while creating a rich evidence trail with an exact evidence score of ≥ 0.99 . This transparent, evidence-grounded system is a step towards making more trustworthy systems for clinical use.

Index Terms—information extraction, neuro-symbolic, evidence verification, natural language processing

I. INTRODUCTION

Clinical notes in electronic health records (EHRs) are among the most valuable sources of medical information available, recording diagnoses, medications, medical investigations and more in rich narrative form. This form is both a strength and a weakness. Patient demographics are routinely collected in structured forms for administrative purposes, and this structured form makes them easy to use in clinical research. Information about the patient’s diagnoses and treatment outcomes often exist solely in narrative form in hospital discharge summaries or letters from clinicians. This makes them useful to deeply understand an individual patient, but very difficult to use to build a broad understanding of a patient population.

Epilepsy is one of the most common neurological diseases, affecting over [x]m people globally. Clinicians use the type of epilepsy, type of seizures and frequency of seizures to determine the severity of the patient’s disease. [1] Clinical trials typically use seizure freedom as their target outcome. [2] The only place this information is routinely recorded is in clinic letters from an epilepsy specialist to the patient’s primary care doctor after the patient visits the clinic. To gather this information at scale, experts are required to complete manual chart review: review each letter, interpret the key details and translate them into a standard form. This process is costly, time-consuming, and error prone. [3] Making this information available in a structured form that you can query in a database to summarise, group and filter can make large-scale research efforts viable.

To make this type of structured data available, there are two broad approaches: collect the data in a structured form or process unstructured data into that form. It is simple to collect medication in a more structured form by using separate fields for medication, dose, unit and frequency, rather than writing ‘Topiramate 100 mg BD’. Applying that to seizure frequency is more difficult because they are not simple facts to record in a single, consistent form. A typical example is, *“focal impaired-awareness seizures have been occurring more often, 2 times per week, whereas previously they only happened once or twice a month.”* Imprecise language like this is known as hedging[5], and paradoxically, stripping it down to a single number would make it less precise and less useful. A system that enables clinicians to record these richer observations and then processes them into a standardised form is ideal.

Natural language processing (NLP) and the subdomain of information extraction (IE) offers a solution for this. Rules-based methods have successfully extracted information on medications and diagnoses, but struggled with the imprecise language and diverse forms of seizure frequency statements. ‘2 to 3 seizures a week’, ‘frequent bouts of seizures’ and ‘4 since her last visit’ are all valid descriptions of the same fact. Large language models (LLMs) have demonstrated significant improvements in semantic understanding than prior methods and have successfully extracted seizure frequency information, but they occasionally hallucinate facts. Combining the semantic understanding of an LLM with the transparency of a rules-based system could provide the most accurate, trustworthy results.

Effective IE can be broken down into three key stages: extracting relevant candidate facts, encoding those facts into a particular form, and selecting which facts to keep and which to filter out. A clinic letter can contain multiple descriptions of seizures, some are historical events while others are current, some are certain seizures while others may be “absence-like episodes”, some are describing the same event but in another context. Seizure frequencies can be expressed as a rate, a vague quantity and a duration. A transparent system would allow you to see not just the final answer, but which steps were taken to achieve it.

This paper evaluates the ideal combination of LLMs and rules across those three stages: extraction, encoding and selection. It was evaluated on two sets of annotated epilepsy clinic letters that reflect two common clinical IE tasks: producing a single label to determine a patient’s current status, and producing a broad inventory of clinical facts, past and present. The paper demonstrates that a modular system combining

LLMs and rules can extract information with comparable accuracy to past methods while preserving a richer evidence chain. The system is available as an open source package at [INSERT LINK](#). Figure 1 displays the web app illustrating some of that chain from evidence to final answer.

II. LITERATURE REVIEW

Epilepsy clinic letters contain facts about medication, diagnoses, investigations and seizure frequency, often covering multiple time periods and with many hedged [?] observations. Fonferko-Shadrach *et al.* treat the letter as an inventory of supported facts across prescriptions, diagnoses, investigations and frequency [?], [?]. Xie *et al.* [?], Holgate *et al.* [?], and Gan *et al.* [?] treat it as one current seizure-frequency state. The decisions about which facts to extract, how to represent them and which facts to keep in the final representation are often bundled into one single task definition. This paper demonstrates that breaking them up into discrete stages with explicit subjective choices can make the system more transparent and flexible.

Wang *et al.* define clinical IE as the automatic extraction and encoding of clinical information from text [?]. Fu *et al.* separate mention detection from concept encoding: the quoted span is distinct from the coded concept [?]. Xie *et al.* extract a frequency sentence and then apply rules to encode it into a number [?]. In much of the remaining epilepsy literature encoding is bundled into the extraction phase, with the model extracting e.g. '4 per month' from a raw statement 'about once a week'. When rules are used to format the answers, they are often described as simple post-processing steps for scoring instead of distinct stages in the pipeline. Those paper shows that recognising extract and encode as separate stages allows you to preserve more of the evidence while still being able to encode it into the required form for downstream use.

INSERT A REVIEW OF EVALUATION PERFORMANCE

Chiticariu *et al.* show that academic papers tend to focus exclusively on benchmark accuracy, while practitioners tend to value transparency, configurability and efficiency at the cost of some accuracy. Fu *et al.* show that this translates to the clinical setting too, and model interpretability is also a critical factor in successful deployment in clinical settings. Clinicians are highly trained in using deductive logic to draw conclusions and they expect systems to expose transparent, interpretable evidence that they can reliably use to make decisions that demand high accountability. Fonferko-Shadrach *et al.* expose evidence spans as a consequence of using a rules-based system, and Gan *et al.* utilise evidence spans as a training signal. This paper presents a transparent evidence chain from exact text to final answer as part of the core output to improve interpretability for downstream users.

Most papers present results of a system trained or tuned on one local corpus. Decker *et al.* show that a highly tuned system on one set of epilepsy letters can perform significantly worse on a set from another epilepsy centre, with seizure-frequency F_1 dropping from 0.82 in-house to 0.40 at a second centre [?]. This is further compounded by the fact that because

medical data is highly sensitive, most papers report results on private datasets that cannot be reproduced or evaluated on other systems. Fonferko-Shadrach *et al.* and c are the two exceptions in this domain. This paper is the first to assess a method on two distinct tasks using two public datasets.

III. METHODS

A. Data

Two datasets were used in this paper. Both datasets are collections of epilepsy clinic letters which record a UK patient's consultation with an epilepsy specialist, often a neurologist or specialist nurse. [?], [?] The NHS requires that clinic letters are sent to the patient's GP after all outpatient visits such as these. [?] They are often the only record of a patient's diagnoses, treatments and outcomes in neurological disorders, and they are written in narrative form. Both datasets are synthetic letters written in part by neurology specialists and modelled on real clinic letters. The letters were annotated by trained researchers and clinicians.

Dataset 1, produced by Fonferko-Shadrach *et al.*, annotates a broad inventory of entities across 9 clinical categories. This paper focuses on the four most important categories - medication, diagnoses, investigations and seizure frequency facts - which were also the focus of the authors' prior paper. The distinguishing feature of this dataset is both the breadth of categories covered and the depth of coverage within them. Each category can include multiple entities per letter, and each entity includes multiple attributes. Figure 1 illustrates an example of this, with multiple medications, diagnoses and investigations. An individual diagnosis mention then includes the attributes DiagCategory, CUI and CUIPhrase. The CUI is the concept unique identifier assigned to each concept in the Unified Medical Language System (UMLS) [?].

The second dataset, produced by Gan *et al.* [?], annotates a single label representing the patient's current seizure frequency burden, often selecting one fact from among many.

Dataset 1 assesses the system's ability to extract a broad inventory of clinical facts and attributes, often with multiple facts per category e.g. multiple kinds of medication, each with its own dose, unit and frequency. Dataset 2 assesses the system's ability to extract a single clinical fact that represents the patient's current seizure frequency burden. This often involves multiple seizure frequency events that need to be aggregated (e.g. 1 in February, 2 in April = 3 per 3 months) or disambiguated (e.g. "She discontinued Valproate on 10 Jul. Shortly afterwards, she experienced 2 to 3 seizures, one triggered by missed medication. She has remained seizure-free since") to produce a single answer.

One dataset is tasked with selecting a single fact, the patient's current seizure burden, from a 400-word letter. The other is tasked with selecting an average of six facts per 200-word letter, covering a broad inventory of clinical facts (medication, diagnosis, investigations, seizure frequency).

Dataset 1 consists of 200 synthetic letters written by trained specialists based on real clinic letters from hospital outpatient consultations with epilepsy specialists in Swansea Bay. The

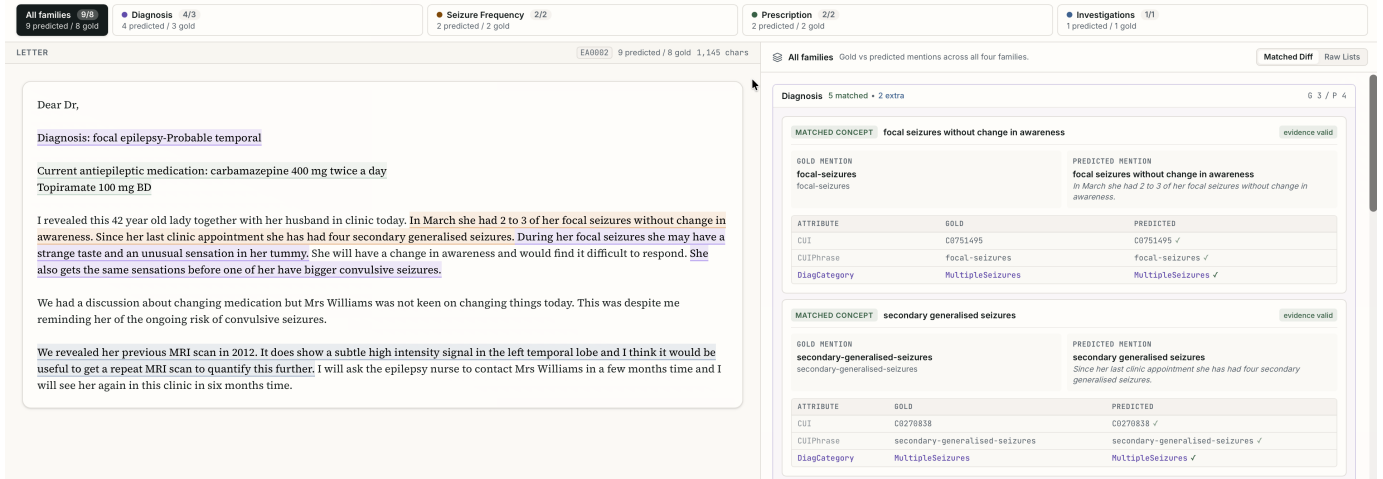


Fig. 1: The left pane is a letter from Fonferko-Shadrach *et al.*'s dataset. Relevant facts about medication, diagnosis, investigations and seizure frequency are highlighted in different colours, reflecting the predictions made in the pipeline. The right pane includes a few diagnoses predictions matched against the gold standard annotations in the source dataset. This illustrates the final outputs of the primary pipeline, and the link between the final answers selected and the raw evidence cited.

letters were double annotated by four trained researchers and clinicians according to strict annotation guidelines. The dataset consists of 9 distinct categories describing a patient's experience of epilepsy. In this paper we focused in the four most important categories: the medication used (drug name, dose, unit, frequency), the diagnosis (type of seizures e.g. tonic-clonic seizures), the investigations used to confirm the diagnosis (e.g. MRI results) and how often the patient has seizures (e.g. focal seizures for around 1 year before this). An example letter is shown in Figure 1a. The dataset was split into

Dataset 2 consists of 1,500 letters using a combination of human and generative AI. Ten synthetic letter templates were created based on real clinic letters from King's College Hospital. GPT-5 generated 500 diverse short descriptions of seizure frequency. Human annotators read and assigned gold-standard labels to each description. The description-label pairs were then inserted into the parameterised templates to create the full dataset of 15,000 letters. This paper used a subset of 1,500 letters provided by the author. The gold-standard annotation consists of a single label describing the patient's current seizure frequency burden, e.g. seizure free for 5 months. An example letter is shown in Figure 2a.

	Dataset 1	Dataset 2
Dataset Size (dev/test)	200 (140/60)	1,200 (750/450)
of Le	Method improves over baseline in 3 key ways	F1, evidence, schema validity, uncertainty
Figure 1: Architecture	Method is modular, transparent, reproducible, and clinically logical	Pipeline stages, representative example
Table 3: Ablation	Deterministic and LLM components are complementary	LLM-only, rules-only, LLM + schema validation, LLM + evidence verifier, LLM + concept normalisation, LLM + temporal resolver, full hybrid
Figure 2: Error Analysis	Deterministic and LLM components have distinct error profiles that you can calibrate for	Hallucinations, low recall, schema validity, evidence coverage, normalisation errors, temporal contradiction, abstention rate

B. Architecture

IV. RESULTS

A. Development Environment

B. Methods

C. Models

D. Structured Evidence

E. Ablations

F. Error Analysis

V. DISCUSSION

VI. CONCLUSION

Entity	Claim it supports	Evidence shown	Reviewer risk
Table 1: Dataset Summary	Method is robust and generalisable	Dataset size, labels, splits, class balance	How valuable is synthetic data?
Table 2: Main Results	Method improves over baseline in 3 key ways	F1, evidence, schema validity, uncertainty	What do schema and evidence validity mean?
Figure 1: Architecture	Method is modular, transparent, reproducible, and clinically logical	Pipeline stages, representative example	Architecture may seem unnecessarily convoluted
Table 3: Ablation	Deterministic and LLM components are complementary	LLM-only, rules-only, LLM + schema validation, LLM + evidence verifier, LLM + concept normalisation, LLM + temporal resolver, full hybrid	Unclear separation between rules and LLM; unclear rules
Figure 2: Error Analysis	Deterministic and LLM components have distinct error profiles that you can calibrate for	Hallucinations, low recall, schema validity, evidence coverage, normalisation errors, temporal contradiction, abstention rate	Difficulty identifying the most meaningful errors

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Please number citations consecutively within brackets [?].
The sentence punctuation follows the bracket [?].

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